

WEYBURN, SK, Canada

WMO: 714520

Lat: 49.700N

Lon: 103.800W

Elev: 589

StdP: 94.45

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.1	(c) -29.3	(d) -35.4	(e) 0.1	(f) -32.0	(g) -32.4	(h) 0.2	(i) -29.2	(j) 15.7	(k) -13.5	(l) 13.8	(m) -12.0	(n) 3.7	(o) 310	(p) 0.734

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	31.7	19.1	29.6	18.6	27.7	17.8	22.1	27.8	20.5	26.3	19.2	25.2	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	16.1	25.1	18.6	14.4	23.5	17.2	13.2	21.8	67.9	28.1	62.0	26.0	57.4	25.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.9	(b) 11.4	(c) 10.2	(e) -36.0	(f) 35.8	(g) 3.4	(h) 2.2	(i) -38.4	(j) 37.4	(k) -40.4	(l) 38.7	(m) -42.3	(n) 40.0	(o) -44.8	(p) 41.6		
(a) 12.9	(b) 11.4	(c) 10.2	(e) -36.0	(f) 35.8	(g) 3.4	(h) 2.2	(i) -38.4	(j) 37.4	(k) -40.4	(l) 38.7	(m) -42.3	(n) 40.0	(o) -44.8	(p) 41.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-13.7	-12.2	-5.0	4.2	11.0	16.2	19.5	18.6	13.1	5.0	-4.0	-11.2
	DBStd	13.34	8.51	8.20	8.02	5.90	4.55	3.57	3.14	3.54	4.64	5.22	6.92	8.02
	HDD10.0	3318	736	621	466	185	43	2	0	0	21	169	419	657
	HDD18.3	5535	994	854	724	424	231	82	23	40	166	414	669	916
	CDD10.0	963	0	0	1	11	74	189	294	267	114	13	1	0
	CDD18.3	138	0	0	0	0	3	18	59	48	9	0	0	0
	CDH23.3	1927	0	0	0	8	86	260	694	676	196	8	0	0
	CDH26.7	633	0	0	0	1	16	66	237	254	59	1	0	0
Wind	WSAvg	4.8	5.1	4.9	5.3	5.4	5.4	4.7	4.2	4.2	4.6	4.9	4.8	4.8
Precipitation	PrecAvg	391	15	12	16	30	55	67	56	48	44	21	11	16
	PrecMax	578	42	30	50	108	170	233	192	197	110	66	22	34
	PrecMin	252	2	2	0	3	1	14	6	0	6	0	1	4
	PrecStd	88	8	7	12	24	38	45	42	40	27	17	6	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	6.0	15.8	24.2	28.8	31.6	34.3	34.9	31.3	24.3	15.6	6.7
		MCWB	3.0	3.3	8.2	12.8	15.0	19.2	20.8	20.2	17.0	13.5	8.8	3.5
	2%	DB	2.9	3.1	11.0	19.9	25.7	28.5	31.3	32.0	28.5	19.9	10.4	3.9
		MCWB	1.2	1.7	6.1	10.7	14.2	18.3	20.3	18.6	16.2	11.5	5.9	1.9
	5%	DB	0.9	1.0	7.2	17.2	23.2	26.2	29.2	29.5	25.5	16.3	7.3	1.7
		MCWB	-0.2	0.2	4.1	9.2	12.8	16.9	19.8	18.3	15.1	9.7	4.0	0.5
	10%	DB	-1.8	-0.8	4.1	14.2	20.5	24.2	27.3	27.2	22.3	13.3	4.6	-0.6
		MCWB	-2.6	-1.5	2.4	7.7	11.6	16.1	19.0	17.4	13.9	8.3	2.4	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.5	9.2	13.6	17.2	23.2	24.3	22.8	20.0	14.4	9.3	3.7
		MCDB	5.1	5.4	14.6	21.6	24.5	28.9	29.3	30.2	25.6	22.0	15.8	5.8
	2%	WB	1.3	1.7	6.5	11.5	15.3	20.2	22.7	20.9	17.7	12.3	6.3	1.8
		MCDB	2.7	3.0	10.8	18.6	22.1	24.6	28.2	27.4	25.3	18.5	10.0	3.3
	5%	WB	-0.1	0.2	4.2	9.9	14.0	18.4	21.2	19.6	16.1	10.3	4.2	0.3
		MCDB	1.0	0.9	6.7	15.8	20.6	23.7	26.5	26.6	23.4	15.2	6.7	1.6
	10%	WB	-2.4	-1.5	2.4	8.2	12.7	17.2	19.9	18.5	14.6	8.7	2.6	-1.6
		MCDB	-1.7	-0.8	4.1	13.4	19.0	22.3	25.5	25.0	21.0	12.8	4.4	-0.7
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.1	9.7	12.2	13.5	12.0	13.3	14.3	13.8	11.6	9.9	9.4
		MCDBR	11.7	10.6	13.4	17.9	18.1	15.9	16.4	18.2	19.2	17.5	14.0	11.5
		MCWBR	10.3	9.1	8.8	9.8	8.6	7.3	7.4	7.3	8.7	10.0	9.5	9.5
	5% WB	MCDBR	11.6	10.3	12.4	16.7	15.1	13.4	14.2	15.2	16.5	15.8	13.2	10.9
		MCWBR	10.4	8.9	8.5	9.7	7.9	7.3	7.7	7.3	8.2	9.4	9.2	9.3
Clear-Sky Solar Irradiance	taub	0.247	0.270	0.272	0.327	0.340	0.352	0.361	0.373	0.325	0.285	0.254	0.244	
	taud	2.370	2.339	2.507	2.398	2.413	2.422	2.416	2.358	2.464	2.551	2.503	2.403	
	Ebn at Noon	826	885	953	920	918	906	891	862	876	866	820	780	
	Edh at Noon	63	84	86	107	111	111	110	111	89	68	55	53	
All-Sky Solar Radiation	RadAvg	1.39	2.47	3.79	4.93	5.83	6.12	6.64	5.30	3.97	2.38	1.37	1.01	
	RadStd	0.13	0.22	0.30	0.33	0.43	0.40	0.36	0.38	0.36	0.27	0.19	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.17	-1.38	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page